Program : **Diploma in Robotic Process Automation**	
Course Code : **5309C**	**Embedded and IOT System Design Lab**
Semester: **5**	Credits: **1.5**
Course Category: **Programme Elective**	
Periods per week: **3 (L: 0 T: 0 P: 3)**	Periods per semester: **45**

Course Objectives:

- Learn to understand and perform the fundamental programming concepts and data manipulation of modern 8051 microcontroller.
- To understand and analyze to interface simple peripheral devices to a Microcontroller.
- To study and understand IOT, their characteristics of components and basic awareness of Arduino / Raspberry Pi
- To obtain practical knowledge about the security aspect into the IoT design in real world problems.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic knowledge of Digital Design and Computer Architecture	3302	Digital Design and Computer Architecture,	3
Basic functions and features of computer, operating system and internet applications, basic programming skills in Python	1008	Introduction to IT system lab	1

Course Outcomes:

On completion of the course student will be able to:

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	To Understand the features of 8051 microcontroller for performing data manipulation in assembly or C programming language using 8051 trainer kit.	12	Applying
CO2	Analyze and to have an in-depth knowledge on interfacing the external devices to the controllers.	12	Applying
CO3	To understand and analyze IoT, Raspberry Pi and also able to install software setup of Arduino/ Raspberry Pi.	12	Applying
CO4	Understand the vision of IoT from a global context for secure use of Devices, Gateways and Data Management	6	Applying
	Series Test	3	

CO – PO Mapping:

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	3	2	2	3		1	2
CO2	3	2	2	3		1	2
CO3	3	2	2	3		1	2
CO4	3	2	2	3		1	2

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**To Understand the features of 8051 microcontroller for performing data manipulation in assembly or c programming language using 8051 trainer kit**		

M1.01	Understand and perform Assembly language programming of Addition / subtraction / multiplication / division of 8/16 bit data.	3	Applying
M1.02	Demonstrate and perform Assembly language programming of Largest/smallest from a series and Sum of a series of 8 bit data.	3	Applying
M1.03	Demonstrate and perform Assembly language programming of Sorting of data in Ascending/Descending order.	3	Applying
M1.04	Understand the concept and perform Assembly language programming of Square / cube / square root of 8 bit data.	3	Applying
CO2	**Analyze and to have an in-depth knowledge on interfacing the external devices to the controllers**		
M2.01	Interfacing experiments using 8051 Trainer kit and interfacing modules with Display (LED/Seven segments/LCD)	3	Applying
M2.02	Keyboard interfacing with 8051 trainer kit	3	Applying
M2.03	Stepper motor interfacing with 8051 trainer kit	3	Applying
M2.04	Time delay generation and relay interface.	3	Applying
	Series Test – I	1.5	
CO3	**To understand and analyze IoT, Raspberry Pi and also able to install software setup of Arduino/ Raspberry Pi**		
M3.01	Familiarization with the concept of IOT, Arduino Raspberry Pi and perform necessary software installation	4	Applying
M3.02	Calculate the distance using a distance sensor and basic LED functionality, Calculate temperature using a temperature sensor using Raspberry Pi	4	Applying
M3.03	Calculate the distance using a distance sensor and basic LED functionality, Calculate temperature using a temperature sensor using Node MCU	4	Applying

CO4	**Understand the vision of IoT from a global context for secure use of Devices, Gateways and Data Management**		
M4.01	To understand and explore various types of Sensors	2	Applying
M4.02	Develop and implement applications using Arduino and Raspberry Pi	2	Applying
M4.03	Explore Open Source tools for Security and Privacy issues in IoT.	2	Applying
	Series Test – II	1.5	

**** - Micro Project**

Students are expected to do a micro project in modern 8051 microcontroller during the course for the purpose of continuous evaluation. This experiment shall be included in the bona-fide record. Example: Develop program such as

· Program to rotate a stepper motor 50^0 in the clock wise direction

· Program to generate a square wave of 1 kHz with duty cycle 33%. Use timer 1 in interrupt mode with a crystal frequency of 11.0592 MHz.

· Program to toggle pin P1.4 every second using interrupts for a frequency of 22 MHz. Use timer 1 in mode 1

. Interfacing program for Realization of Boolean expression through port.

. Develop a system using existing IoT platforms

. Implement the working of temperature sensor using IoT

Text / Reference

T/R	Book Title/Author
T1	Muhammed Ali Mazidi & Janice Gilli Mazidi, R.D. Kinley, The 8051 microcontroller and Embedded System, Pearson Education, 2nd edition
T2	Embedded Systems – Architecture Programming and Design by Raj Kamal, 2nd edition, McGraw Hill.
T3	Vijay Madisetti and Arshdeep Bahga, “Internet of Things (A Hands-on-Approach)”, 1st Edition, VPT, 2014

T4	Francis daCosta, “Rethinking the Internet of Things: A Scalable Approach to Connecting Everything”,1st Edition, Apress Publications, 2013.
T5	Cuno Pfister, Getting Started with the Internet of Things, O‟Reilly Media, 2011, ISBN: 978-1- 4493-9357-1

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/108102169
2	https://nptel.ac.in/courses/117104072
3	https://github.com/connectIOT/iottoolkit
4	https://www.arduino.cc/
5	http://www.zettajs.org/

List of Experiments:

1. Write an Assembly language program to perfom Addition / subtraction / multiplication / division of 8/16 bit data .
2. Write an ALP program to find Largest/smallest from a series .
3. Write an ALP program to find Sum of a series of 8 bit data.
4. Write an ALP program for Sorting (Ascending/Descending) of data.
5. Write an ALP program Square / cube / square root of 8 bit data.
6. Write an interfacing program of Display (LED/Seven segments/LCD)
7. Write an interfacing program of keyboard interface with 8051 trainer kit
8. Write an interfacing program of Stepper motor interfacing with 8051 trainer kit
9. Write an interfacing program of Time delay generation and relay interface
10. Familiarization with the concept of IOT, Arduino Raspberry Pi and perform necessary software installation
11. Calculate the distance using a distance sensor and basic LED functionality, Calculate temperature using a temperature sensor using Raspberry Pi
12. Calculate the distance using a distance sensor and basic LED functionality, Calculate temperature using a temperature sensor using Node MCU

13. To understand and explore various types of Sensors
14. Develop and implement applications using Arduino and Raspberry Pi
15. Explore Open Source tools for Security and Privacy issues in IoT